

EEE8089

M2M Technology IoT

Coursework, Part 1

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Aims

- Part 1 of the practical coursework
 - network architecture
 - access from home
 - design problems to solve

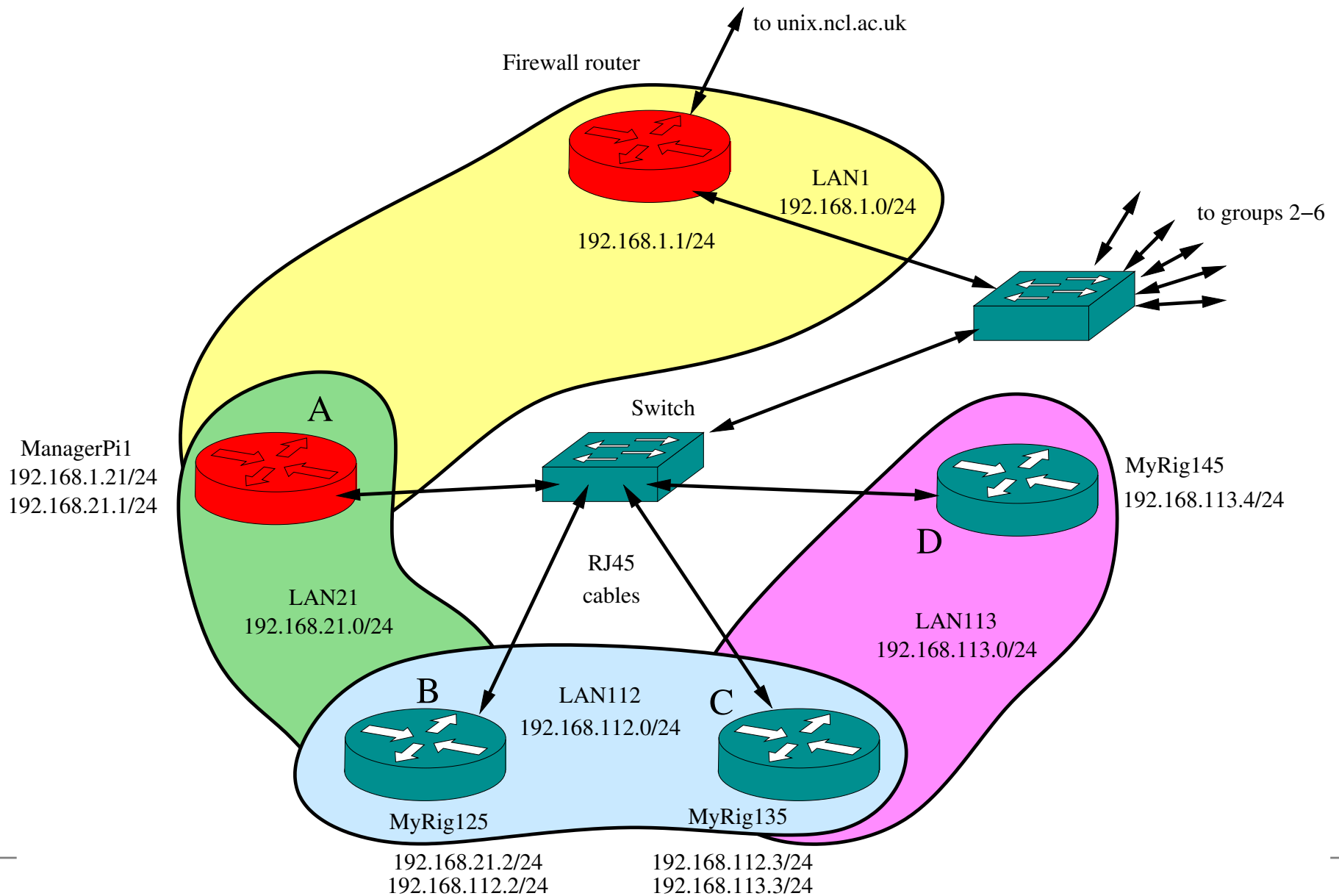
(Part 2 in a separate lecture)

Three Architectures

- Target Architecture – the main design effort
- Instrumental Architecture – pre-configured to provide access to the components of the design
- Access From Home Architecture – a scheme for accessing the above two configurations remotely
- Four RPi 3B+ devices, replace # with group number, multiple names for the devices with several OS images
 - ManagerPi#
 - MasterRig#2x, MyRig#25, MyRig#26, MyRig#27, MyRig#28
 - MasterRig#3x, MyRig#35, MyRig#36, MyRig#37, MyRig#38
 - MasterRig#4x, MyRig#45, MyRig#46, MyRig#47, MyRig#48

Note: the last three are often referred to as RigPi

Target Architecture



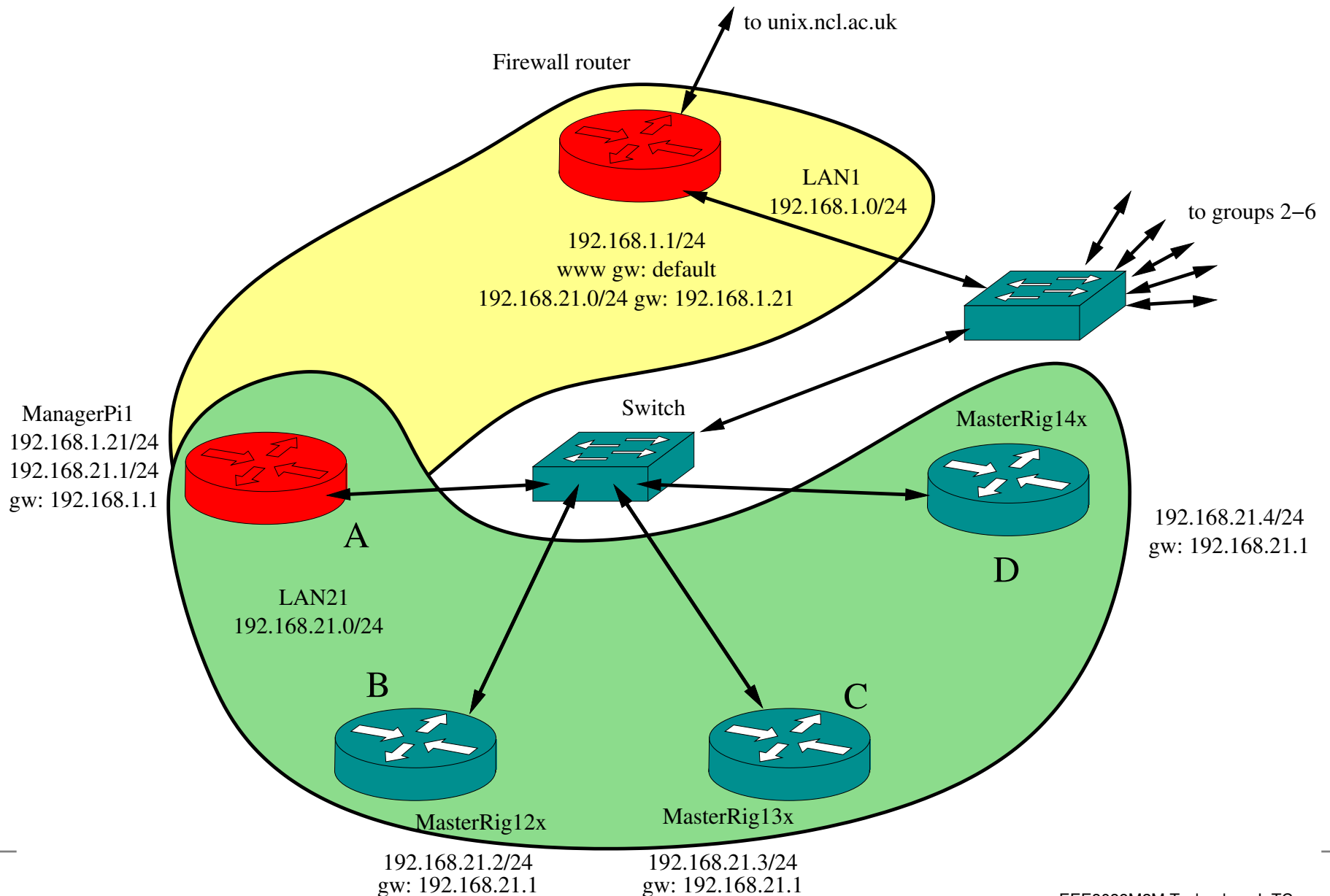
IP Addresses for Groups

Group	ManagerPi#	MyRig#2i	MyRig#3i	MyRig#4i
1	192.168.21.1	192.168.21.2	192.168.112.3	192.168.113.4
		192.168.112.2	192.168.113.3	
2	192.168.22.1	192.168.22.2	192.168.122.3	192.168.123.4
		192.168.122.2	192.168.123.3	
3	192.168.23.1	192.168.23.2	192.168.132.3	192.168.133.4
		192.168.132.2	192.168.133.3	
4	192.168.24.1	192.168.24.2	192.168.142.3	192.168.143.4
		192.168.142.2	192.168.143.3	
5	192.168.25.1	192.168.25.2	192.168.152.3	192.168.153.4
		192.168.152.2	192.168.153.3	
6	192.168.26.1	192.168.26.2	192.168.162.3	192.168.163.4
		192.168.162.2	192.168.163.3	

Tasks to Complete

- Find the IP addresses, define the LANs and determine which gateways to use.
- Configure the devices, test them, debug any problems.
- Test the connectivity: each device should be able to talk to any other and to www.
- Use tcpdump to examine Ethernet, IP and TCP packets, find the key fields in the headers of these packets
- Examine the latency and throughput of connections within one LAN, between two LANs and between three LANs
- Generate and distribute ssh keys between the devices, so ssh sessions could be started without typing a password. Experiment with forwarding a data pipeline through ssh between two devices.

Instrumental Architecture



Using the Instrumental Architecture

- Get ssh access to ManagerPi# by using Access From Home Architecture (described below).
- From ManagerPi access RigPi devices – they are running in MasterRig mode.
 - Each MasterRig device can be logged in as user pi with provided password
 - The OS images of MyRig (four spares per device) are mounted as ~/rootfs5...8
 - Edit the chosen OS image, then reboot RigPi into this image (MyRig mode) by creating a file ~/boot_5 (or 6,7,8) and then rebooting the device
 - The next reboot will take you back into the MasterRig mode

Instrumental Architecture IP Addresses

	ManagerPi#	MasterRig#2x	MasterRig#3x	MasterRig#4x
Group 1	192.168.21.1	192.168.21.2	192.168.21.3	192.168.21.4
Group 2	192.168.22.1	192.168.22.2	192.168.22.3	192.168.22.4
Group 3	192.168.23.1	192.168.23.2	192.168.23.3	192.168.23.4
Group 4	192.168.24.1	192.168.24.2	192.168.24.3	192.168.24.4
Group 5	192.168.25.1	192.168.25.2	192.168.25.3	192.168.25.4
Group 6	192.168.26.1	192.168.26.2	192.168.26.3	192.168.26.4

Hardware Reboot of MyRig

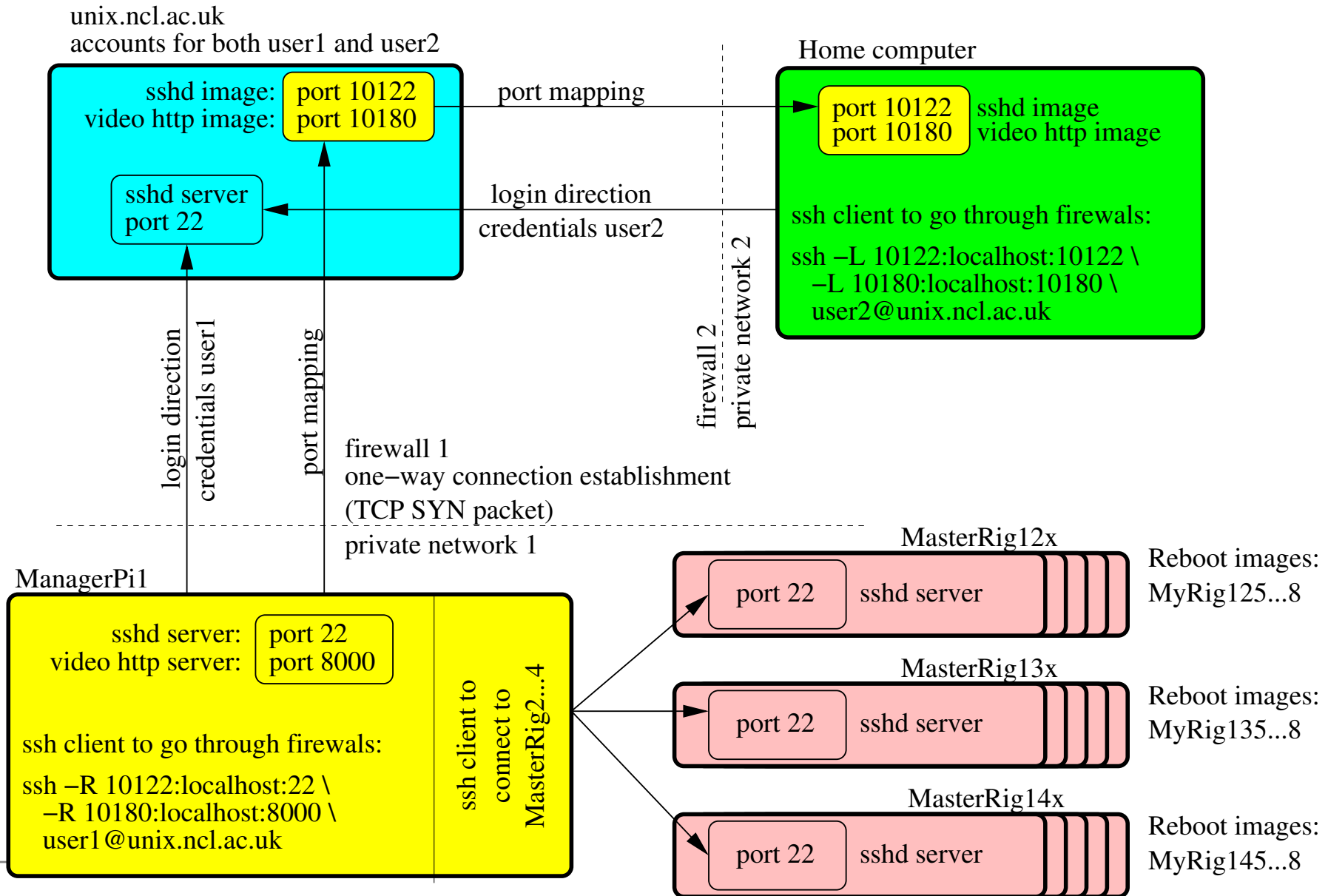
- If network connection to the MyRig fails, then hardware reboot is possible by issuing a command from ManagerPi
- Soft reboot (for MyRig#2x device)
 - `echo 0 > /sys/class/gpio/gpio17/value`
 - `echo 1 > /sys/class/gpio/gpio17/value`
- Hard reboot (may cause filesystem corruption – try to avoid)
 - `echo 0 > /sys/class/gpio/gpio27/value`
 - `echo 1 > /sys/class/gpio/gpio27/value`

MyRig#2i: soft reboot gpio17, hard reboot gpio27

MyRig#3i: soft reboot gpio22, hard reboot gpio10

MyRig#4i: soft reboot gpio9, hard reboot gpio11

Access From Home Architecture



Using from Home

Everywhere below replace # with your group number

- Set up the ssh tunnels to import the ports from unix.ncl.ac.uk
 - Group #: 10#22, 10#80
- Login into ManagerPi#
 - `ssh -Y -p 10#22 login_name@localhost`
- View the video feed in the browser (tested with Firefox)
 - `http://localhost:10#80`